

A Decision-Level Fusion Framework for Enhanced Multi-Class Chronic Wound Detection

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Abstract—Chronic wounds pose a significant global health challenge, affecting millions and placing a heavy financial burden on healthcare systems. Accurate classification of wound types is critical for effective diagnosis and treatment planning, as it enables clinicians to select the most appropriate care strategies. While several diagnostic tools exist, most current approaches are limited to binary classification and struggle to accurately differentiate among multiple wound types. A robust multi-class classification system is essential to improve diagnostic accuracy and reduce the time and cost associated with wound care. This research proposes a smart diagnostic framework using decision-level fusion to enhance the performance of multi-class foot ulcer detection, aiming to support clinicians with faster and more reliable wound assessment. We trained four pre-trained models VGG19, DenseNet201, MobileNetV2, and NASNetMobile—on an augmented dataset of 14,640 images. To improve prediction accuracy, we implemented decision-level fusion techniques such as hard voting, soft voting, maximum likelihood, and stacking. The stacking model using XGBoost achieved the highest accuracy of 89.47%, significantly outperforming individual models.

Index Terms—Chronic wound, Multi-class, decision fusion, transfer learning, Stacking Ensemble

I. INTRODUCTION

A wound is defined as a disruption in the integrity of the skin or underlying tissues, typically caused by physical, chemical, or biological agents. Wounds are generally categorized into two types: acute and chronic. Acute wounds, such as surgical incisions or minor lacerations, follow a predictable healing trajectory and usually close within a few weeks [1]. In contrast, chronic wounds fail to progress through the normal phases of healing in a timely manner, often persisting for several months or even years [1], [2]. These chronic wounds impose a substantial clinical and economic burden, contributing to increased rates of hospitalization, prolonged treatment durations, and heightened risk of complications, including infection and amputation. It is estimated that chronic wounds affect approximately 1% to 2% of the population in developed nations [1]–[3].

Recent advances in machine learning (ML) have demonstrated significant potential in the domain of automated and objective wound evaluation, particularly for chronic wounds such as diabetic foot ulcers, venous leg ulcers, and pressure injuries. Deep learning (DL) especially convolutional neural networks (CNNs) has been widely adopted for tasks such as wound image classification [2]–[9], segmentation [5], [10]–[12], tissue classification [13], and healing prediction [14]. CNNs are capable of learning complex spatial patterns in medical images, enabling robust identification of wound boundaries, differentiation of tissue types (e.g., necrotic, granulation, slough), and assessment of wound severity over time.

Such intelligent systems not only improve diagnostic accuracy, but also reduce clinician workload and enable real-time decision support, thereby contributing to improved patient outcomes [4], [15]. In addition to single-model architectures, ensemble learning, transfer learning, and feature fusion techniques have been explored to further enhance model performance and generalization, especially when working with limited or imbalanced datasets. Previous research [2]–[6], [12], [15]–[17] has investigated various deep learning architectures, including ensemble methods and feature fusion techniques, to enhance the performance of wound classification models.

However, most existing studies [4]–[7] have focused primarily on binary classification tasks such as distinguishing between ulcerated and non-ulcerated wounds or have been limited to a narrow range of classes, typically between two and four. As a result, the development of comprehensive multi-class wound classification systems remains relatively underexplored. Moreover, while widely adopted architectures such as ResNet and EfficientNet have been extensively studied in this context. The performance and applicability of other advanced CNN models including VGG19, NASNetMobile, MobileNetV2, and DenseNet201 have not been sufficiently evaluated, particularly on the AZH chronic wound dataset. This research makes the following key contributions to the field of Wound care and clinical decision support:

- Evaluate the effectiveness of state-of-the-art pre-trained convolutional neural network (CNN) models specifically VGG19, NASNetMobile, MobileNetV2, and DenseNet201 for the task of multi-class classification of chronic wound images, including diabetic foot ulcers, pressure ulcers, surgical wounds, and venous ulcers, using the AZH dataset.
- Investigate whether fine-tuning these CNN models on the AZH dataset for the first time can yield superior classification performance compared to existing approaches.
- Enhance the overall diagnostic accuracy of the wound classification system through the application of decision-level fusion strategies, such as weighted fusion, maximum likelihood estimation, and hard voting.
- Improve the robustness and generalizability of the classification framework by incorporating ensemble stacking techniques involving machine learning models, including XGBoost, Logistic Regression, and Random Forest.

II. RELATED WORK

This section discusses the existing approaches for the classification of chronic wound types. The existing approaches for wound type classification are mainly focused on binary classification; only a few studies have performed the multi-class classification.

CNNs have become the foundational architecture in the development of wound classification systems. Aldoulah et al. [2] proposed a fused multi-class CNN model demonstrating notable performance on diverse wound types. The framework consists of two EfficientNet-B4 models, which differ in terms of layer freezing and the activation function used. Specifically, two models were employed and fused via concatenation to extract the more informative features. For the 6 classes of the AZH dataset, they obtained 83.19% accuracy. Similarly, Rostami et al. [3] developed an ensemble-based classifier that combines patch-wise and image-wise CNN predictions to enhance classification performance. The ensemble consists of two AlexNet classifiers for multi-class wound image classification. One classifier extracts image-level information, while the other processes patch-level information using the sliding window technique. The combined outputs from both classifiers are then fed into a Multilayer Perceptron (MLP) to predict the class label. However, the process of patch generation and processing in the patch-wise classifier adds complexity to the ensemble model, increasing both time and computational cost. For the 3-class classification (diabetic, pressure, and venous-arterial), average accuracies of 82.9% and 87.7% were achieved on the Medetec and AZH datasets, respectively. This proposed method [3] demonstrates acceptable accuracy for classifying venous and surgical wounds, but it shows the lowest accuracy when classifying pressure and diabetic wounds. Classifying diabetic foot ulcers (DFUs) accurately is a critical step toward early intervention and improved patient outcomes. To address the challenges posed by limited labeled datasets and high inter-class similarity, Rehman et al. [6] proposed DFU-SIAM, a novel Siamese neural network (SNN) architec-

ture designed specifically for DFU image classification. The model employs an ensemble of EfficientNet and Bidirectional Encoder Representation from Image Transformers (BEiT) as the backbone for the two identical branches (or “twin” networks) of the Siamese structure. This combination leverages the strength of EfficientNet’s feature extraction capabilities and BEiT’s contextual representation power, allowing the model to effectively learn subtle differences between ulcer types. The proposed approach achieved a classification accuracy of 95% across four distinct diabetic foot ulcer categories, outperforming several existing deep learning baselines.

Sarp et al. [18] classified chronic wounds into four categories: diabetic, lymphovascular, pressure injury, and surgical using the pre-trained VGG16 architecture. The authors reported an average precision and recall of 83% and 71% respectively on the test set, indicating satisfactory performance. Importantly, they incorporate the XAI technique for interpretability of the model’s predictions, thereby improving transparency and trust in the automated wound classification process. Recent advancements in medical image analysis have focused on increasing both accuracy and interpretability in automated diagnosis systems. Ahmed et al. [7] introduced XAI-FusionNet, leverages multi-scale feature fusion to better represent complex ulcer morphology, while integrating explainable AI techniques such as Grad-CAM to provide interpretable visual feedback. To construct FusiNet, combines three high-performing CNNs DenseNet201, VGG19, and NASNetMobile for diabetic foot ulcer classification. Each model contributes unique strengths: VGG19 captures fine-grained features due to its depth, NASNetMobile offers efficiency and scalability, and DenseNet201 improves feature reuse through dense connections. This multi-architecture fusion allows the network to learn diverse feature representations, resulting in a high classification accuracy of 99.01% on a binary classification task. [19] demonstrated that the fusion of handcrafted machine learning (ML)-based low-level features with deep learning (DL)-based high-level features extracted via convolutional neural networks (CNNs) yields significantly better performance than using single feature set. The combined features of handcrafted low-level and CNN based feature fusion approach with Logistic regression classifier. It has given the best set of results with precision of 95.55dataset.

Moreover, Munadi et al. [8] demonstrated the effectiveness of decision fusion across thermal and RGB modalities for DFU detection, reaching 84.6% accuracy. This study proposed a deep learning framework that combines thermal image analysis with a decision fusion strategy to detect early-stage diabetic foot conditions. The model incorporates multiple CNN classifiers (MobileNetV2 and ShuffleNet) whose outputs are aggregated at the decision level, improving the robustness and accuracy of the final diagnosis. In [9], a multi-modal deep neural network was proposed that utilizes both wound images and anatomical location data for more accurate classification. Experimental results demonstrated that incorporating wound location improved classification accuracy across multiple chronic wound categories. It achieved the highest

classification accuracy of 83.04% across six wound categories (BG, N, D, P, S, A + V). The most effective configuration was a hybrid architecture combining VGG19 for image feature extraction and a Long Short-Term Memory (LSTM) network for sequential location encoding, demonstrating the benefit of fusing visual and contextual information in wound classification tasks.

Our research builds on the principle of leveraging complementary model strengths by employing four pre-trained convolutional neural networks VGG19, DenseNet201, MobileNetV2, and NASNetMobile as independent classifiers. Each model is selected for its unique architectural advantages and trained to generate class probabilities for multiclass chronic wound classification, distinguishing between six clinically relevant categories: diabetic, venous, pressure, surgical, natural (healthy), and background. Rather than concatenating feature embedding, we adopt a decision-level fusion strategy, combining the outputs of these models to make final predictions. This ensemble approach captures diverse decision patterns across models, enhancing robustness and classification performance. Although our work focuses solely on RGB images, the publicly Advancing the Zenith of Healthcare (AZH) dataset comprising 10,798 augmented images across the six categories provides a solid foundation for evaluating decision fusion strategies in the context of multiclass chronic wound diagnosis.

III. MATERIAL AND METHODOLOGY

A. Dataset

The dataset used for this study is sourced from a publicly available chronic wound dataset AZH Dataset. The AZH dataset was collected over a two-year clinical period at the AZH Wound and Vascular Center in Milwaukee, Wisconsin. It comprises 730 wound images in .jpg format. The images vary in size, with widths ranging from 320 to 700 pixels and heights ranging from 240 to 525 pixels [9]. It contains foot images representing different categories of chronic wounds, including diabetic foot ulcers, pressure ulcers, venous ulcers, surgical wounds, as well as healthy foot images. Additionally, background images are included to represent non-foot objects, aiding in reducing false positives during classification. This dataset was carefully selected for its diversity in chronic wound types and its applicability in real-world clinical scenarios. Sample of Wound images are shown in Fig 1.

B. Data Pre-Processing

The initial dataset images varied in size and quality. As the pre-trained models used in this study require fixed input dimensions, all images were resized to a standard resolution of $224 \times 224 \times 3$ pixels. This resolution maintains the necessary visual details while being computationally efficient. Additionally, the images were normalized by scaling pixel values to the range of $[0, 1]$ to accelerate convergence during training. Data augmentation techniques including Random Rotation, Horizontal and Vertical Flipping, Random Zoom and Brightness Adjustment were applied to artificially expand the



Fig. 1: Wound Images

dataset by generating new images with slight modifications. This not only increased the dataset size but also introduced variations in the data, which improved the model's ability to generalize to unseen images. After augmentation, the dataset results in a total of 10,798 images significantly used for model training and testing as shown in Table I.

TABLE I: DATASET DESCRIPTION

Wound Type	No. of Original Images	No. of Augmented Images
Diabetic Foot Ulcers	185	400
Pressure Ulcers	134	736
Venous Ulcers	247	400
Surgical Wounds	164	544
Healthy Foot Images	100	672
Background Images	100	992

C. Proposed Methodology

In this work, we employed four state-of-the-art pre-trained deep learning models, each contributing unique architectural strengths to the ensemble. VGG19, DenseNet201, MobileNetV2, and NASNetMobile were selected based on their proven effectiveness in large-scale image classification tasks and their strong generalization capabilities across diverse datasets. VGG19 uses a deep yet straightforward architecture with 19 layers, making it easy to implement and fine-tune. Despite its age, it remains a robust baseline for image classification tasks. Dense connections between layers allow the DenseNet201 model to reuse features, improving learning efficiency and reducing redundancy. MobileNetV2 has fewer parameters and low computational cost that enhance speed and performance without sacrificing accuracy. NASNetMobile offers a strong balance between accuracy and computational efficiency, making it suitable for mobile and cloud-based medical systems. In this research, decision-level fusion techniques were employed to improve the performance of individual deep learning models for chronic wound classification. Decision-level fusion aggregates predictions from multiple independently trained classifiers. The ensemble strategies used in our work include Majority Voting, Weighted Soft Voting, Maximum Likelihood Fusion, and Stacking with Meta-Model learning. The proposed methodology is shown in Fig 2.

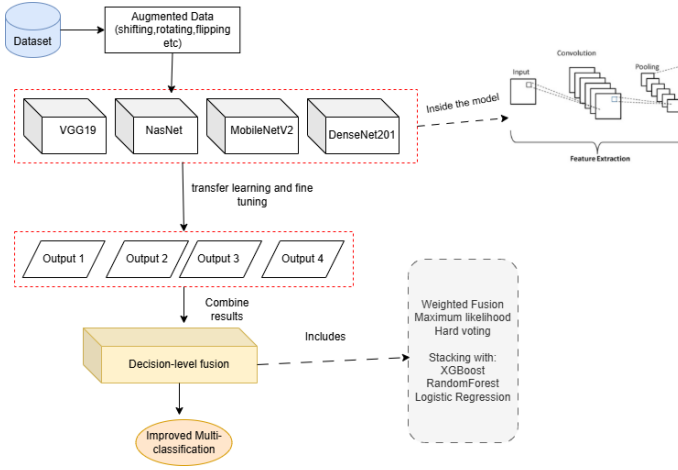


Fig. 2: Block Diagram of the Proposed Methodology

D. Decision Fusion

The Decision Fusion (DF) technique offers several notable advantages. It enables the effective integration and management of heterogeneous data sources, allowing for the efficient combination of information from multiple origins. DF also aids in the reduction of dimensionality and the suppression of various types of noise [20], which collectively contribute to enhanced model performance. Moreover, it plays a critical role in reducing ambiguity and uncertainty, thereby improving the robustness and reliability of analytical outcomes.

In high-stakes domains such as healthcare, the decision-making process must be carried out with a high degree of accuracy to minimize potential errors and ensure improved quality of care. By supporting more precise and data-driven decision-making, DF contributes to better clinical outcomes and the overall effectiveness of healthcare systems [21].

In this research, decision-level fusion techniques were employed to improve the performance of individual deep learning models for the classification of chronic wounds. Decision-level fusion aggregates predictions from multiple independently trained classifiers. The ensemble strategies used in our work include Majority Voting, Weighted Soft Voting, Maximum Likelihood Fusion, and Stacking with Meta-Model learning.

1) *Maximum Likelihood Fusion*: It is a probabilistic ensemble strategy that combines the outputs of multiple classifiers by assuming conditional independence among them. Choose the class label that maximizes the joint likelihood of all classifiers' outputs, under the assumption that each classifier provides a class-conditional probability estimate.

2) *Weighted Soft Voting Fusion*: In this approach, the softmax probability distributions from each model are combined using a weighted sum. Higher weights are assigned to models that perform better in the validation data. The final predicted class is the one with the highest weighted average probability across all classifiers.

3) *Hard Voting*: This is the most basic ensemble method where each model votes for a predicted class, and the class

receiving the majority of votes is selected as the final prediction.

4) *Stacking*: It is an advanced ensemble learning technique that leverages the predictions of multiple base models as input features for a higher-level classifier, known as the meta-model. The stacking process in our study follows these key steps:

- 1) The original dataset is divided into training and validation sets to ensure unbiased evaluation.
- 2) Four deep learning architectures VGG19, DenseNet201, MobileNetV2, and NASNetMobile are individually trained on the training set.
- 3) Each base model generates predictions on the validation set in the form of softmax probabilities or class labels. These predictions are then combined to form a new dataset, which serves as input features for the meta-model.
- 4) A meta-classifier is trained on this new dataset to learn the optimal way to integrate the outputs of the base models and improve overall classification performance.

In our experiments, three types of meta-models were evaluated:

- **Logistic Regression**: A linear classifier that is well-suited for learning from the probabilistic outputs of base models.
- **Random Forest**: An ensemble of decision trees capable of capturing complex, nonlinear relationships among the base models' predictions.
- **XGBoost**: A gradient boosting framework that offers strong predictive power through sequential tree learning.

E. Transfer Learning Model

The concept of transfer learning is particularly useful for addressing challenges related to insufficient datasets during the training process. By leveraging knowledge gained from pre-trained models on large, diverse datasets, transfer learning allows for effective feature extraction and model initialization on new, smaller datasets. This approach significantly reduces training time and computational resources while often improving model generalization and accuracy. Transfer learning has proven especially valuable in domains where labeled data is scarce or expensive to obtain, such as medical imaging, natural language processing, and other specialized fields.

In this work, we employed four state-of-the-art pre-trained deep learning models, each contributing unique architectural strengths to the ensemble. VGG19, DenseNet201, MobileNetV2, and NASNetMobile were selected based on their proven effectiveness in large-scale image classification tasks and their strong generalization capabilities across diverse datasets. Each pre-trained model was loaded using weights learned from ImageNet. The models were initialized without their final classification layers (fully connected layers). VGG19 uses a deep yet straightforward architecture with 19 layers, making it easy to implement and fine-tune. Despite its age, it remains a robust baseline for image classification tasks. Dense connections between layers allow the DenseNet201 model to reuse features, improving learning efficiency and reducing redundancy. MobileNetV2 has fewer parameters and

low computational cost that enhance speed and performance without sacrificing accuracy. NASNetMobile offers a strong balance between accuracy and computational efficiency, making it suitable for mobile and cloud-based medical systems.

IV. EXPERIMENTAL SETUP

The fused feature vectors were passed through a dense neural network for final classification. The classifier consisted of:

- Input layer with the fused feature vector. Two hidden layers with 512 and 256 neurons using ReLU activation
- Dropout layers (rate = 0.5) to prevent overfitting
- Output layer with softmax activation for multi-class classification

Experimental studies were conducted on chronic wound images obtained from the AZH dataset, which includes six wound classes: diabetic, pressure, venous, surgical, and others. The dataset was split into two parts: 80% of the data was used for training, while 20% was used for testing. In this research work, the experiments were performed on Intel® core™ i9-10850K at CPU speed 3.6 GHz with 64 GB RAM including an NVIDIA 3060 GPU with 12 GB memory. Operating System was Windows 10 Pro and system type is 64-bit operating system. The Adam optimizer was initially used with a learning rate of 0.001 during the training phase for its adaptive learning rate capabilities and faster convergence. Categorical Cross-Entropy was used as the loss function, and Adam was selected as the optimizer with an initial learning rate of 0.001 followed by batch size of 32 for 20 epochs.

V. RESULTS AND DISCUSSION

The performance of the individual CNN models (VGG19, DenseNet201, MobileNetV2, and NasNetMobile) and the ensemble fusion methods (weighted soft voting, hard voting, maximum likelihood fusion, and stacking) was evaluated using standard classification metrics: Accuracy, Precision, Recall (Sensitivity) and F1-score. These metrics were chosen to provide a comprehensive evaluation of the models across all six chronic wound classes.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (2)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (3)$$

$$\text{F1 Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4)$$

VGG19 achieved an overall accuracy of 80%, demonstrating strong performance across most classes. It performed particularly well on BG, N, and D with F1-scores above 0.95, indicating high precision and recall. Moderate performance was observed for S and V, while P showed relatively lower accuracy. Moreover, DenseNet201 obtained an overall

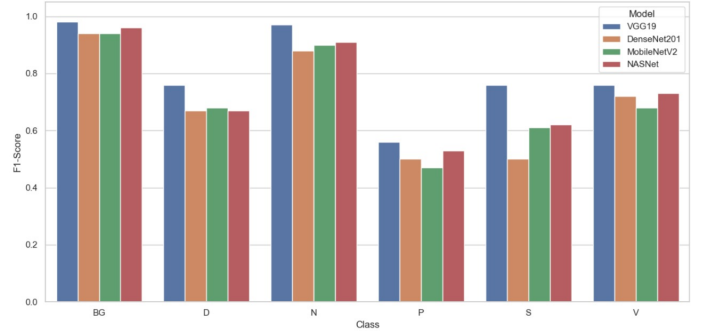


Fig. 3: Comparison of F1 scores for individual classifiers across six classes

accuracy of 72%, showing relatively strong performance for classes BG, N, and V, with F1-scores above 0.70. The model performed exceptionally well on class N (F1-score: 0.88) and BG (F1-score: 0.94). However, performance on classes P and S was lower (F1-score: 0.50 each), suggesting challenges in predicting these classes accurately. Fig 3 shows F1-Score for individual classifiers across all AZH dataset classes.

Fig 4 presents a comparative analysis of various proposed fusion methods, including Maximum Likelihood, Weighted Soft Voting, Hard Voting, and Stacking. Among these approaches, the XGBoost-based stacking ensemble demonstrated the best overall performance, achieving the highest accuracy of 89.47% and a macro F1-score of 89%, consistently outperforming other fusion strategies across all wound classes. This was closely followed by the Random Forest (88.72% accuracy) and Logistic Regression (84.71% accuracy), both of which also exhibited strong classification capabilities, particularly in challenging classes such as P, S, and V.

Although Weighted Soft Voting achieved a respectable accuracy of 82.96%, its performance in minority classes particularly class P (F1-score: 0.60) was notably less effective compared to the stacking methods. In contrast, simpler fusion techniques such as Maximum Likelihood and Hard Voting yielded lower overall performance, especially in classes P and S. These results suggest that such basic fusion strategies may be insufficient for capturing the complexity and variability inherent in the outputs of individual models.

Figure 4 shows the comparative results of different proposed fusion method using VGG19, MobileNet and NasNet. Among all methods, the XGBoost stacking ensemble achieved the best performance with the highest overall accuracy 89.47%, average precision 92.83% and macro F1-score 89.3%, consistently outperforming others across all classes. It was followed closely by Random Forest (88.72%) and Logistic Regression (84.71%), both showing strong performance on difficult classes like P, S, and V. While, weighted soft voting obtained an accuracy of 82.96%, it was less effective than stacking methods in minority classes like P (F1: 0.60). Max Likelihood and Hard Voting showed lower performance overall, especially in classes P and S, suggesting that simple fusion techniques may not fully capture the diversity across model outputs.

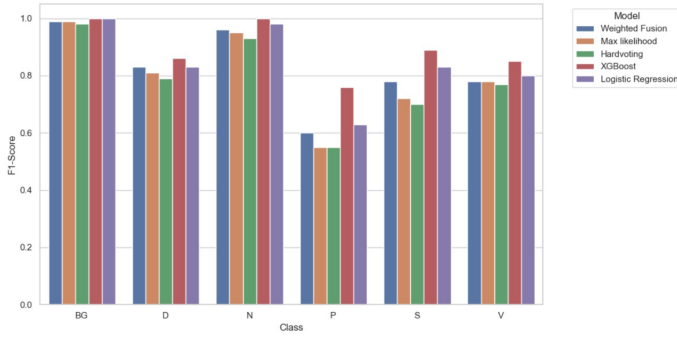


Fig. 4: Comparison of F1 scores for Decision fusion approaches across six classes

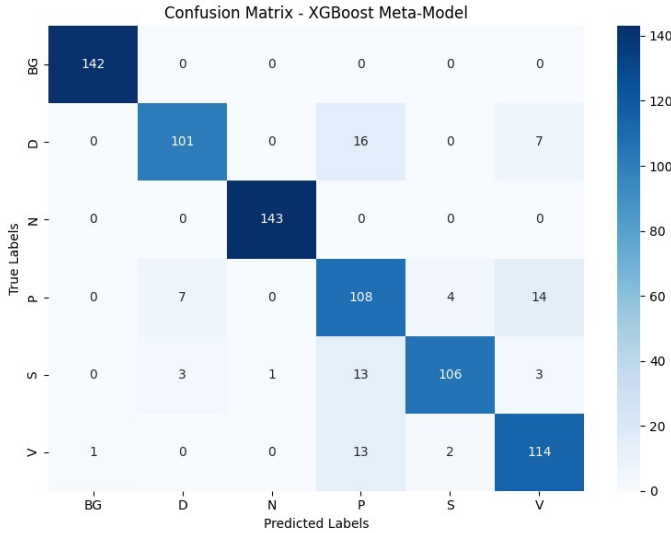


Fig. 5: Confusion Matrix of XG boost Classifier

Figure 5 shows the confusion matrix of the best performance stacking model i.e XGBoost.

Table II summaries the performance evaluation of wounds classification using convolution neural network. Several studies have explored the use of deep learning models for chronic wound and diabetic foot ulcer (DFU) classification, with varying success across classification types, datasets, and model architectures. Zaid et al. [2] addressed a relatively complex six-class chronic wound classification task using the AZH dataset and achieved 83% accuracy with EfficientNetB4. This performance is notable given the increased difficulty of multiclass classification, whereas inter-class similarity and data imbalance often hinder model accuracy. In comparison, Rostami et al. [3] applied an ensemble strategy combining patch-wise and image-wise classification pipelines using AlexNet and a multilayer perceptron, achieving only 68.7% accuracy for a four-class problem on the same dataset. This lower performance may stem from reliance on a relatively shallow network like AlexNet and a more limited fusion strategy, highlighting the importance of both depth and integration design in ensemble methods.

In binary classification, the model's task is inherently simpler, higher accuracies are typically observed. Shuvo et al. [7] focused on DFU detection and applied a feature fusion method using DenseNet201, VGG19, and NASNetMobile. They achieved 83% accuracy, reflecting strong performance on visual image classification despite variability in lesion presentation. Khairul et al. [8], on the other hand, achieved 100% accuracy using thermal images and decision-level fusion of ShuffleNet and MobileNet. The remarkable performance can be attributed to the complementary nature of thermal imaging, which captures temperature-based physiological differences. Furthermore, their decision fusion strategy capitalizes on agreement between classifiers to suppress misclassifications, suggesting that modality selection and decision integration are critical factors in DFU detection accuracy.

Overall, these findings highlight that while binary classification of DFUs may achieve good performance using optimized pipelines and specialized modalities, multiclass chronic wound classification remains a greater challenge due to the visual similarities between wound types and class imbalance issues. Our proposed approach, applied within the more challenging multiclass domain, demonstrates competitive performance and highlights the effectiveness of ensemble architectures and decision-level fusion in addressing such complexities.

VI. CONCLUSION

Chronic wounds such as diabetic, venous, pressure, and surgical ulcers pose a significant burden on healthcare systems and require timely, accurate identification for effective treatment. This study aims to develop a reliable AI-based system capable of classifying wound types into six categories: Background, Diabetic, Pressure, Venous, Surgical, and Natural (healthy skin). Initially, we fine-tuned and evaluated three pre-trained convolutional neural network (CNN) architectures VGG19, MobileNetV2, and NASNetMobile on an augmented dataset comprising over 14,800 images. The models demonstrated the following classification accuracies: VGG19 achieved 80%, MobileNetV2 achieved 72%, and NASNetMobile achieved 74%. Each model demonstrated distinct strengths in feature representation due to architectural differences, yet their standalone performance was limited in capturing the complex inter-class variations among wound types. To improve the classification accuracy, we investigated a decision-level fusion strategy. Instead of combining features, we fused the final softmax predictions of the three models using a majority voting scheme. This ensemble method enhanced the system's robustness and reduced model specific biases and misclassifications. The fused model, when integrated with XGBoost as a meta-classifier, achieved an improved classification accuracy of approximately 89.47%, significantly outperforming the individual models. Evaluation metrics such as precision, recall, and F1-score further confirmed the reliability and generalizability of the proposed fusion-based classification system for chronic wounds.

To improve model generalization and minimize bias, it is recommended to collect a larger and more diverse dataset

TABLE II: Comparison of the Proposed Techniques with State-of-the-art Studies of Wound Classification

Studies	Dataset	No of Classes	Model	Accuracy
Zaid et al. [2]	AZH RGB images	Multiclass (D,V,S,P,N,BG)	EfficientNetB4	83%
Rostami et al. [3]	AZH RGB images	Multiclass (D,V,S,P,N,BG)	AlexNet	68.7%
Shuvo et al. [7]	DFU RGB images	Binary Diabetic Foot ulcer	Feature Fusion: Densenet201	98.2%
Khairul, et al [8]	Thermograms Dataset	Binary Diabetic Foot ulcer	VGG19,NASNetMobile	100%
This Study	Augmented AZH RGB images	Multiclass (D,V,S,P,N,BG)	Decision Fusion	80%
			ShuffleNet MobileNet	74%
			VGG19	72%
This Study	Augmented AZH RGB images	Multiclass (D,V,S,P,N,BG)	NasNet	81.05 %
			MobileNetV2	82.96 %
			Decision Fusion	80%
			Maximum Likelihood	84.71%
			Weighted Soft Voting	88.72 %
			Hard Voting	89.47 %
			Logistic Regression	
			Random Forest	
			XGBoost	

of wound images. Inclusion of data from varied populations, skin tones, and clinical environments will ensure better representation and robustness of the model across real-world applications. Also, to enhance clinical trust and facilitate interpretability, future work should integrate explainability methods such as Gradient-weighted Class Activation Mapping (Grad-CAM) and SHapley Additive exPlanations (SHAP).

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